Qijin Xu

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Lead AI/ML Architect

Summary

Lead AI/ML Architect with 5+ years building production AI and the data systems beneath it. Architected an LLM-powered RAG assistant that resolved about 90% of questions on its own for 50,000+ users and returned 525% ROI within 12 months (CGI Impact Award 2024). Now leading a LangGraph agent for a proprietary programming language, projected to cut manual work by 50%. Earlier, built a fraud loss-ratio model with a 321% lift over a top-five bank benchmark.

Experience

2025 — **Lead AI/ML Architect, AI Center of Excellence**

CGI Inc. · Toronto, ON · Jan 2025 – present

- Lead an AI agent for a proprietary programming language: LangGraph dual-state dynamic workflow, vLLM-served LLMs with RAG, EBNF structured outputs, and PII filtering. It writes documentation, extracts business rules, maintains legacy scripts, and prepares them for migration. Projected 50% less manual work.

2023 — **Data Architect & Engineering Lead, Senior Consultant**

CGI Inc. · Toronto, ON · Aug 2023 – Dec 2024

- CGI Impact Award 2024: architected and shipped an LLM-powered enterprise RAG assistant for insurance support (LangChain; Azure, AWS, GCP). About 90% autonomous resolution for a 50,000+ user base; 525% ROI within 12 months. Product page: cgi.com/canada/en-ca/insurance-chatbot
- Built an OpenLineage-based lineage tool that auto-documents multi-source data flows, with LLM-assisted root-cause suggestions. Reduced investigation time across about 3 incidents per month.
- Automated AxiomSL rule updates at RBC with an LLM: 80% less manual updating, governance controls preserved.

2021 — **Data Engineer, Consultant**

CGI Inc. · Toronto, ON · Aug 2021 – Jul 2023

- Managed Canada's largest automobile-insurance policy-history database (2 TB, SQL Server): 99.9% uptime, median queries under 200 ms.
- Built production ETL/ELT pipelines for 22+ micro data marts (under 6 tables each, monthly SLAs) for finance, analytics, and delivery teams. Report runtime cut from 30 to 5 minutes.
- Shipped a gradient-boosting and logistic-regression loss-ratio model, batch-scored in a Python/T-SQL pipeline: 321% lift in recovered fraud losses versus a top-five bank benchmark.

2020 — **Research Assistant, Computer Vision (part-time)**

Elder Laboratory, York University · Toronto, ON · May 2020 – Aug 2021

- Worked on AttentiveVision's computer-vision pipeline for hockey broadcast video (PyTorch, TensorFlow, OpenCV). Trained a CNN-LSTM stack reaching mAP ≥ 70% on event detection across 18 hours of game video.
- Built real-time player tracking with a Kalman filter and OpenPose: under 100 ms latency at 30 FPS, for live speed and heat-map analytics.
- Built an event-detection annotation loop (goals, timeouts, and penalties) with active learning, raising label reliability to Cohen's kappa ≥ 0.92.

2019 **Full-Stack Developer Intern**

Nascent Digital · May 2019 – Aug 2019

- Built features in React, TypeScript, and Node.js, including major parts of the company homepage rewrite, and shipped TELUS product pages with UX, QA, and DevOps, design to deploy in 7 days.
- Introduced visual-regression testing (Nightwatch.js, Puppeteer), removing about 90% of manual QA checks.

Skills

Languages	Python, Java, Go, SQL (T-SQL, PostgreSQL), TypeScript/JavaScript, C++
AI/ML	PyTorch, JAX, LangGraph, RAG, agent workflows, Model Context Protocol (MCP), LLMs, vLLM, OpenAI API, FAISS, Pinecone
Data and streaming	Apache Spark, Kafka, Airflow, OpenLineage, Redis
Cloud and systems	Azure, GCP, AWS, Docker, Kubernetes, Terraform, microservices, gRPC, CI/CD

Education

York University, Toronto, ON. B.Sc. Computer Science, Specialized Honours, 2017 – 2021.

GPA 3.9/4.0 · Dean's List

Awards and certifications

CGI Impact Award (2024), for the enterprise RAG assistant

CGI Impact Award (2026)

Microsoft Certified: Azure Solutions Architect Expert (2022)

Google Cloud Professional Cloud Developer (2023)

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